

With this perspective in mind, which neurochemical system is the most crucial for controlling the balance between more automatic and more deliberate cognitive processing? Based on previous [research](#), my colleagues and I had a hunch that the serotonergic system might be a good place to look. Serotonin is released into the brain from multiple different nuclei, [including](#) the raphe nuclei and the medullary reticular formation. Once released, serotonin has an extremely complex set of effects, as there are a wide range of different serotonin receptors that can turn the presence of the neurochemical into different cellular outcomes. In fact, researchers have struggled to produce simplified explanations for the system that capture all of its inherent complexity.